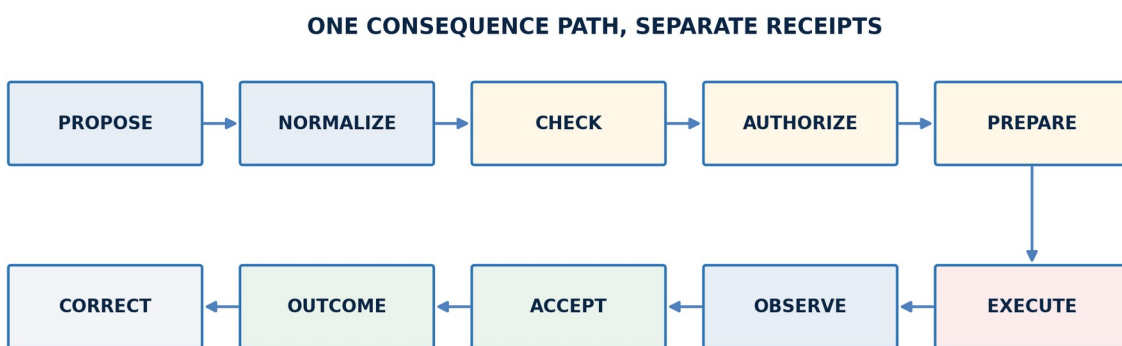


State Transition Protocol

A framework architecture for controlled state transitions in agentic systems

The State Transition Protocol (STP) is a framework architecture for moving from generated proposals to scope-bound accepted state. It separates evidence, authority, execution, observation, acceptance, and outcome. Its accepted-state ledger preserves the event history and current projection.



A failed gate records a refusal and preserves the prior accepted projection.

Figure 1. The consequence path uses a separate receipt for each control function.

Core claim: A model can propose a change. A tool can apply a change. Accepted state advances only through a declared projection rule after the required evidence, authority, observation, and conflict checks.

Released for independent review and reproducible testing.

Abstract

Agentic AI research often assigns new labels to recurring functions: proposal, memory, guard, delegation, execution, observation, commit, and review. The State Transition Protocol (STP) defines a function-first crosswalk and a control architecture that keeps those functions independently checkable. Its accepted-state ledger is an append-only event record plus a deterministic, policy-versioned projection of accepted state. Accepted state means the latest state recognized within a declared scope by a declared authority. World state, observed state, candidate state, and desired state remain distinct. The framework adds typed deltas, evidence and authority gates, prepare and commit semantics, post-action observation, causal ordering, conflict rules for parallel agents, operational mode transitions, and outcome review. Its deterministic claim is narrow: identical valid events, ordering rules, schemas, and policy versions produce the same accepted projection. External tools, sensors, people, and physical systems can remain stochastic or faulty. Security follows a zero-trust model across models, tools, adapters, witnesses, authorities, and ledger operators. Physical safety equations apply only when their system models and assumptions are valid. The framework is designed for conformance testing through replay, conflict, authorization, postcondition, tamper-evidence, and recovery fixtures. It is a versioned synthesis of established distributed systems, formal methods, control theory, provenance, and current agent research. Candidate contributions include the accepted-state distinction, a functional equivalence test for agentic lexicons, and a unified change-control sequence for harness-neutral use.

TL;DR PTE: The protocol keeps an AI project from turning one persuasive answer into operating reality. It records what was suggested, what evidence supports it, who permitted it, what changed, what a witness saw, what state was accepted, and whether the result worked.

Claim boundary

Scope	Meaning
The paper provides	A theoretical control model, shared vocabulary, event semantics, equations, test fixtures, and revision process.
A deployment must provide	Domain rules, legitimate authority, trustworthy observations, secure enforcement, recovery, and accountable operation.
The deterministic scope	Ledger validation and state projection under pinned versions and ordering rules.
The empirical scope	Tool effects, model behavior, witness quality, human decisions, outcomes, and physical dynamics.
Novelty status	Candidate synthesis. Literature convergence supports relevance. Evaluation and independent review remain required.

AI assistance disclosure

AI tools assisted literature retrieval, draft construction, diagram production, document formatting, and deterministic checks. The author retains responsibility for claims and decisions. Publication details appear at the end of the paper.

Contents and reading path

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Fast path: Read Sections 2, 3, 4, 7, and 11 for the operating model. Read Sections 12 and 13 for the research comparison and accepted revisions. Appendix C records the architecture decisions.

1. The lexicon problem

TL;DR PTE: Research teams frequently rename the same control role. A shared function test can preserve useful differences while collapsing cosmetic differences.

Agentic systems contain recurring roles. A component proposes. Another component checks. An authority permits. A tool acts. A witness observes. A record preserves the event. A projection decides which state is currently in force. Papers may call these roles planners, critics, guards, verifiers, orchestrators, memories, checkpoints, traces, or ledgers.

Labels alone cannot establish equivalence. Two components share a functional role only within a declared scope and only when their read set, write set, decision right, evidence burden, failure behavior,

timing, and guarantee surface align. Delta Atlas calls the stable center the function statement. This paper expresses that center as a testable signature [U03, U04].

```
signature(term) = <reads, writes, output, authority, evidence,
                  failure semantics, timing, guarantee, assumptions>
equivalent_scope(a,b) = reviewed_match(signature(a), signature(b), scope)
```

Mapping status	Required interpretation
ALIAS	Signatures match for the declared scope and guarantee.
OVERLAP	Some operations match; authority, timing, or guarantees differ.
RELATED	Terms participate in the same workflow with different functions.
CONTESTED	Sources or reviewers disagree about the mapping.
UNKNOWN	Evidence remains insufficient for a responsible mapping.

Research discipline: A repeated label supplies weak evidence. Independent definitions, executable behavior, and failure semantics supply stronger evidence. Every mapping carries sources and review status.

2. Core thesis and Plain Technical English

TL;DR PTE: The model proposes. Deterministic controls decide whether a proposal may reach execution. A witness checks the effect. A versioned rule decides whether the result enters accepted state.

2.1 Core thesis

Agentic reasoning is a stochastic proposal process. Consequential change requires a separate decision boundary with typed inputs, explicit authority, versioned predicates, conflict control, and receipts. The accepted projection provides a compact current view. The append-only evidence archive preserves how that view was produced.

In this paper, protocol means a versioned set of roles, records, gates, and transition rules. Framework architecture means the design can be implemented across different models, tools, and harnesses. The accepted-state ledger is the record and projection component within that architecture.

This structure aligns with current work on persistent archives and fresh contexts, symbolic guardrails, executable specifications, authenticated delegation, stateful interfaces, and deterministic concurrent execution [M01, H01, E02, A01, T01, C01]. The combined specification remains a candidate synthesis.

2.2 Plain Technical English core

Term	Functional meaning
World state	The external condition at a time. Access is partial through observations.
Observed state	What a named witness reports, including time, coverage, uncertainty, and limits.
Candidate state	A proposed interpretation or change awaiting checks.
Accepted state	The latest scope-bound state a declared authority recognizes under a versioned policy.
Desired state	A goal or target. It carries no evidence that the goal exists or was reached.
Typed delta	A structured proposed transition with reads, writes, preconditions, and expected effects.
Evidence	Recorded support for a defined claim within stated coverage and freshness.
Authority	The legitimate and scoped power to approve, deny, defer, or revoke an action.
Execution	The attempt to create an effect through a tool, transaction, or actuator.
Observation	The post-action measurement or report used to compare expected and actual effect.
Acceptance	The policy decision that advances or preserves the accepted projection.
Outcome	The later measured result, including benefit, harm, drift, and unresolved effects.
Receipt	One append-only event with actors, versions, times, links, status, and integrity data.

2.3 Bridge image

A breaker panel separates functions. A switch requests a circuit change. A breaker enforces a limit. A meter observes current. A licensed person controls alterations. An inspection record preserves what was approved and observed. The ledger applies the same separation of duties to agentic work.

3. State model and accepted projection

TL;DR PTE: External reality, witness reports, proposed state, and accepted state are separate records. The ledger preserves their differences instead of compressing them into one narrative.

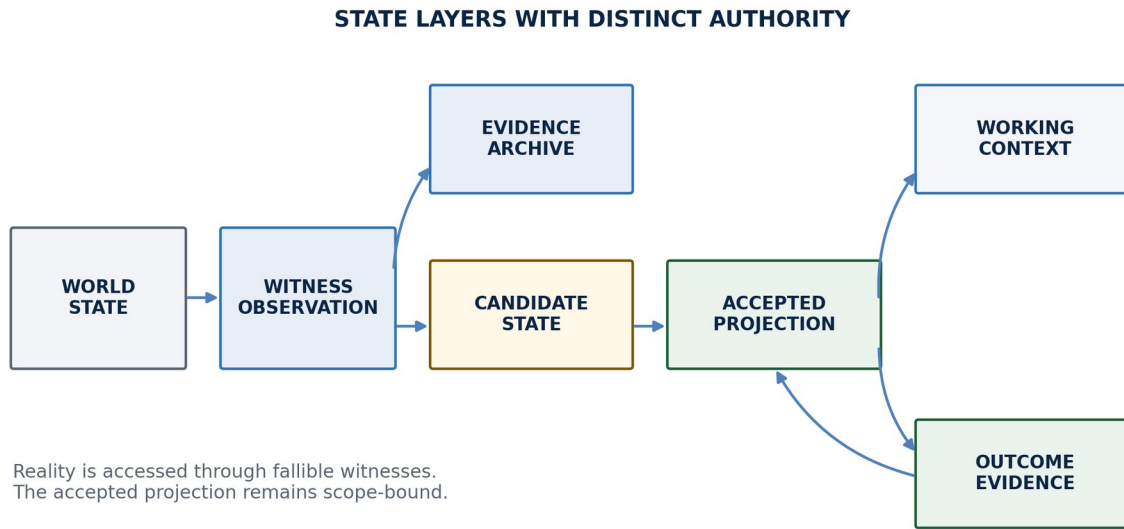


Figure 2. Accepted state is a scope-bound projection over receipts.

3.1 Partial observation

Let $W(t)$ denote external world state. A witness i produces an observation at time t_i :

$$O_i = H_i(W(t_i), \eta_i)$$

H_i witness channel and measurement process
 η_i noise, bias, delay, failure, or adversarial influence

The observation can support a claim within its declared channel and coverage. It supplies no universal view of $W(t)$. Multiple witnesses may disagree because they observe different regions, use different clocks, or share hidden failure modes.

3.2 Deterministic accepted projection

Let L contain accepted ledger events. Let ρ_v denote the pinned projection and policy rules. The current accepted state A_k is reconstructed from the previous accepted state and a ready event e_k :

$$A_{(k+1)} = R(\rho_v)(A_k, e_k)$$

R returns A_k unless e_k is a valid acceptance event for this scope.

The reproducibility claim requires identical event bytes, schemas, policy versions, parent order, and resolver rules. Reproducible projection gives structural determinism. External truth and safe outcomes remain empirical questions [F01, F02, C01].

Accepted means: Recognized within scope by a named authority under a named rule at a named time.
 The label carries no universal truth claim.

4. Transition contract and gate stack

TL;DR PTE: A consequential change moves through prepare, execute, observe, and accept. Each stage can fail independently. One PASS value would hide those differences.

4.1 Typed delta

Group	Minimum content
Identity	delta_id; subject; scope; proposer; correlation_id
Causality	base_version; parent_event_ids; prior_checkpoint
State access	read_set; write_set; expected_version
Transition	operation; proposed_effect; atomicity; idempotency_key
Conditions	preconditions; invariants; evidence_policy; authority_policy
Time	observed_at; prepared_at; expires_at; clock_source
Effect	tool_or_actuator; side_effect_scope; postconditions
Recovery	compensation; rollback_limit; safe_mode; incident_route
Integrity	canonicalization; digest; signatures; adapter_profile

A delta may update several fields atomically when an invariant spans them. The unit of change is the smallest safe transaction, which can exceed one variable.

4.2 Gate stack

Gate	Question answered
Representation	Schema, types, required fields, canonical form, and understood adapter mapping
Evidence	Predicate, witness, coverage, freshness, uncertainty, and common-cause limits
Authority	Identity, delegation, purpose, resource, action, time, consent, and revocation
Invariant	Domain constraints, safe set, policy hierarchy, and operational mode
Concurrency	Base version, causal parents, read/write conflict, lock or compare-and-swap
Commit confirmation	Observed poststate, partial effects, compensation, and acceptance rule

Symbolic guardrails can enforce many requirements when those requirements are stated as checkable predicates. Hong et al. report high symbolic enforceability across studied benchmark requirements. Their result supports explicit predicates within the studied scope [H01].

4.3 Independent status axes

Axis	Allowed values
Evidence	SUPPORTED CONTRADICTED UNRESOLVED UNAVAILABLE INVALID
Authority	APPROVED DENIED DEFERRED ESCALATED REVOKED NOT_REQUIRED
Preparation	READY BLOCKED STALE CONFLICT EXPIRED
Execution	NOT_ATTEMPTED APPLIED PARTIAL FAILED REVERSED
Observation	MATCHED MISMATCHED UNAVAILABLE INVALID
Acceptance	ACCEPTED PRESERVED PROVISIONAL CONTESTED SUPERSEDED EXPIRED
Outcome	MET NOT_MET MIXED TOO_EARLY UNMEASURED

Sequence: Evidence can be supported while authority is denied. Execution can be applied while observation mismatches. A useful outcome can occur after an unauthorized action. The record keeps every statement separate.

5. Functional equivalence across research vocabularies

TL;DR PTE: The crosswalk groups terms by work performed. Each row also names the difference that frequently breaks equivalence.

STP role	Common labels	Equivalence break
Proposed delta	plan; intent; candidate action; proposed tool call	Check whether the object can change state directly.
Working context	scratchpad; context window; short-term memory	Check expiry, authority, and persistence.
Evidence archive	episodic memory; research archive; provenance store	Check source preservation and correction rules.
Accepted projection	checkpoint; current state; operational memory; snapshot	Check authority, conflict, and reconstruction rules.
Check predicate	guardrail; invariant; constraint; policy rule	Check whether enforcement is symbolic, model-based, or advisory.
Witness	verifier; judge; critic; sensor; monitor	Check channel, independence, authority, and error semantics.
STP role	Common labels	Equivalence break
Authority	approval; mandate; delegation; access control	Capability and permission can differ.

STP role	Common labels	Equivalence break
Execution	tool call; transaction; actuation; side effect	A success response can differ from observed effect.
Orchestration	harness; scaffold; runtime; controller	Direction, enforcement, and observation powers can differ.
Receipt	trace; event; audit log; telemetry span	Telemetry can omit authority and state semantics.
Acceptance	commit; confirmation; certification; publish	Commit durability and governance recognition can differ.
Outcome	reward; task success; utility; impact	Measure, horizon, population, and causal strength can differ.

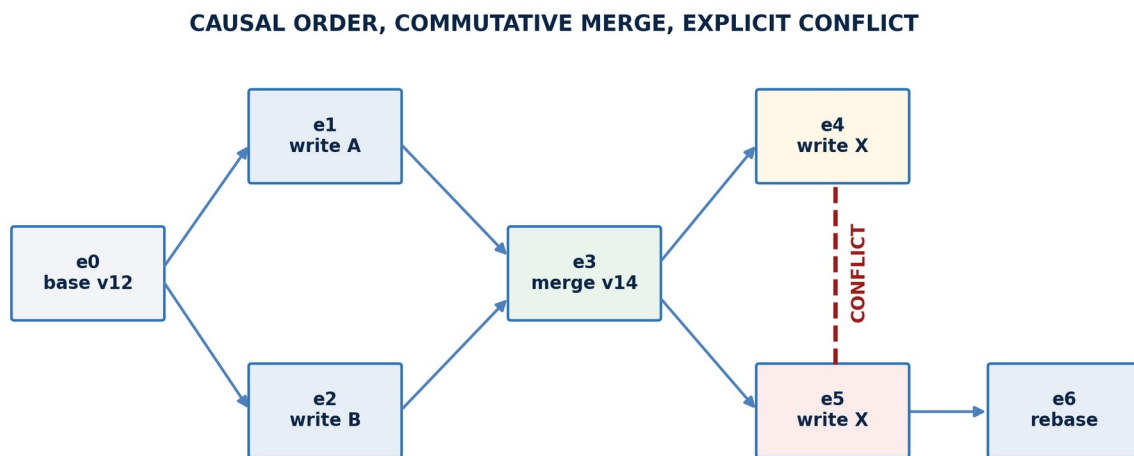
Engram separates a durable Archive from a compact Research Digest. MemoryArena evaluates memory together with later action. MAGE organizes execution state through validated branches. These systems overlap with STP memory roles and use different acceptance semantics [M01, M02, M03].

Better Harnesses, Smaller Models treats instructions, tools, and orchestration as adaptable harness features. STP treats the harness as an implementation surface and keeps the receipt and projection contract stable across harness changes [H03].

Crosswalk warning: Similarity of purpose supports an OVERLAP mapping. An ALIAS mapping also requires matching authority, timing, failure semantics, and guarantees.

6. Parallel agents and concurrent state

TL;DR PTE: Parallel reasoning can improve coverage and speed. Shared state needs causal links, conflict tests, and deterministic merge rules.



Disjoint writes may merge. Competing writes require one winner and a recheck against the new base.

Figure 3. Concurrent deltas require causal parents and explicit conflict behavior.

6.1 Event order

Distributed events can have a partial order. Parent links encode causal dependence. Wall-clock timestamps help investigation and expiry, while causal order and version checks govern state transition. Lamport's happened-before relation provides the foundational distinction [F01].

6.2 Conflict rule

```
conflict(d_i,d_j) =
  W_i intersects (R_j union W_j)
  OR
  W_j intersects (R_i union W_i)
```

Exception: a declared commutative operation with verified merge semantics.

R_i and W_i denote the read and write sets of delta i . Disjoint or verified commutative deltas may merge. Competing noncommutative deltas use a lock, transaction, or compare-and-swap. A stale loser returns for re-evaluation against the new base [F02, F03].

Conflict control: For every L2 or L3 authoritative commit, the gate checks base version and read/write conflicts. A noncommutative conflict returns CONFLICT or STALE and produces zero accepted effects. The fixture target is zero lost updates.

6.3 Partition and availability policy

Situation	Default behavior
Critical accepted state	Prefer preserved state or safe mode during uncertain coordination. Record availability loss.
Local reversible state	Allow a labeled provisional branch when policy permits. Reconcile with explicit conflicts.
Read-only work	Continue from a pinned snapshot and disclose staleness.
Emergency action	Use narrow break-glass authority, expiry, observation, and retrospective review.

Parallel and recursive agent scaling increase coordination demand [C02, C03]. Dynamic communication topologies add efficiency and robustness tradeoffs [C04]. The accepted-state boundary supplies a shared contract for those changing topologies.

7. Operational modes, phase language, and physical models

TL;DR PTE: Use phase for protocol stages. Use mode for governed operating regimes. Physical equations belong only to systems with a valid dynamical model, sensors, actuators, uncertainty bounds, and timing assumptions.

7.1 Discrete and hybrid dynamics

A physical or cyber-physical system can be represented locally as:

```

x_(k+1) = F_q(x_k, u_k, w_k)
y_k      = H(x_k, v_k)
q_(k+1) = Gamma(q_k, y_k, rho_v)

x  system state      u  selected action
w  disturbance        y  observed output
v  observation error  q  operational mode

```

The ledger can record x estimates, u commands, y observations, q transitions, and rule version ρ_v . Control of the physical process requires a connected, timely, and non-bypassable actuator path.

7.2 Mode transition contract

Mode	Meaning
NORMAL	Validated envelope; ordinary action scope
CONSTRAINED	Reduced action set after warning conditions
HOLD	Preserve current safe state while evidence or authority is unresolved
DEGRADED	Limited function with known unavailable components
RECOVERY	Controlled return after repair and verification
EMERGENCY	Narrow high-priority action under explicit break-glass rules

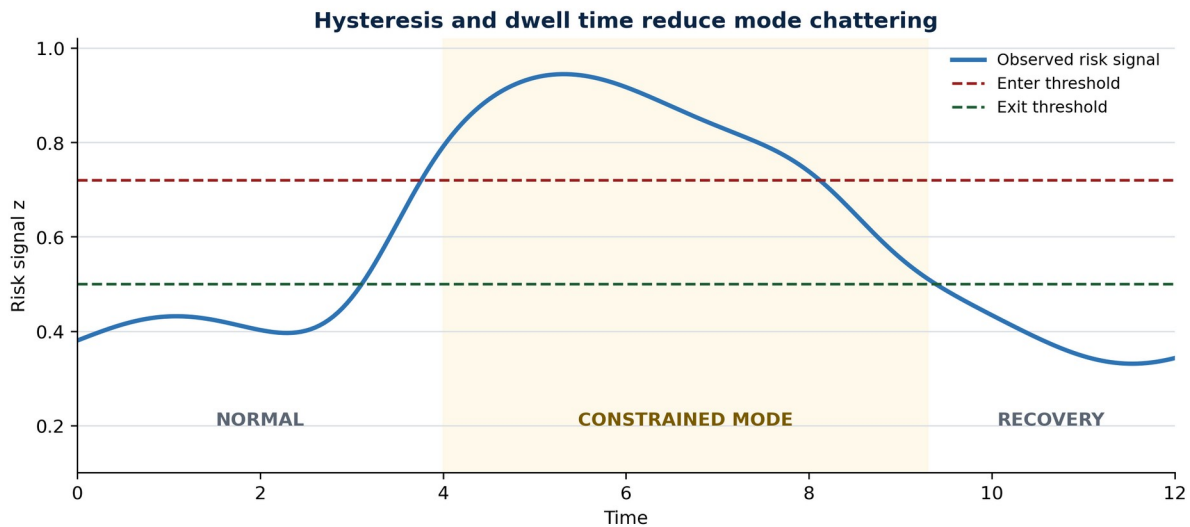


Figure 4. Separate entry and exit thresholds reduce rapid mode switching.

For a risk signal z , enter CONSTRAINED mode after z reaches θ_{enter} for a declared dwell time. Return only after z falls below θ_{exit} for a declared recovery time, with θ_{exit} lower than θ_{enter} . The thresholds require calibration against noise, delay, and consequence.

7.3 Robust safety boundary

For a control-affine system with bounded disturbance d :

$$\begin{aligned}
 &\dot{x} = f(x) + g(x)u + d, \quad d \in D \\
 &C = \{x : h(x) \geq 0\} \\
 &\min_{d \in D} \text{grad } h(x) \cdot [f(x) + g(x)u + d] + \alpha(h(x)) \geq 0
 \end{aligned}$$

Under the applicable control-barrier-function conditions, a selected control u satisfying the robust inequality can keep the safe set C forward invariant. Required assumptions include a suitable model, differentiable h , bounded uncertainty, adequate sensing and actuation, and timely solution of the control constraint [F05, P02, P03].

Composed safety constraints require explicit logical treatment. Combinatorial control barrier functions address nested Boolean and p-choose-r constraints in studied multi-agent systems [P03]. Assume-guarantee contracts support component-level fault diagnosis [P04]. Task-relevant metrics connect perception error to downstream system requirements [P05].

7.4 Language discipline

Phrase	Responsible use
Physical phase transition	Use for a physical model with an order parameter, regimes, and validated dynamics.
Operational mode transition	Use for NORMAL, HOLD, DEGRADED, RECOVERY, or similar governed states.
Protocol phase	Use for propose, check, authorize, execute, observe, accept, and review.
Friction or impedance	Use as a marked design analogy. Supply operational measures separately.
Tipping point	Use with an explicit signal, forecast horizon, uncertainty, and mitigation timing [P06].

Plain example: Boiling water illustrates a physical phase transition. Switching an agentic workflow into HOLD illustrates an operational mode transition.

8. Zero trust and the consequence boundary

TL;DR PTE: Every model, tool, adapter, witness, approver, policy service, clock, and ledger operator can fail or be compromised. Each consequential step requires identity, scope, verification, and a recorded result.

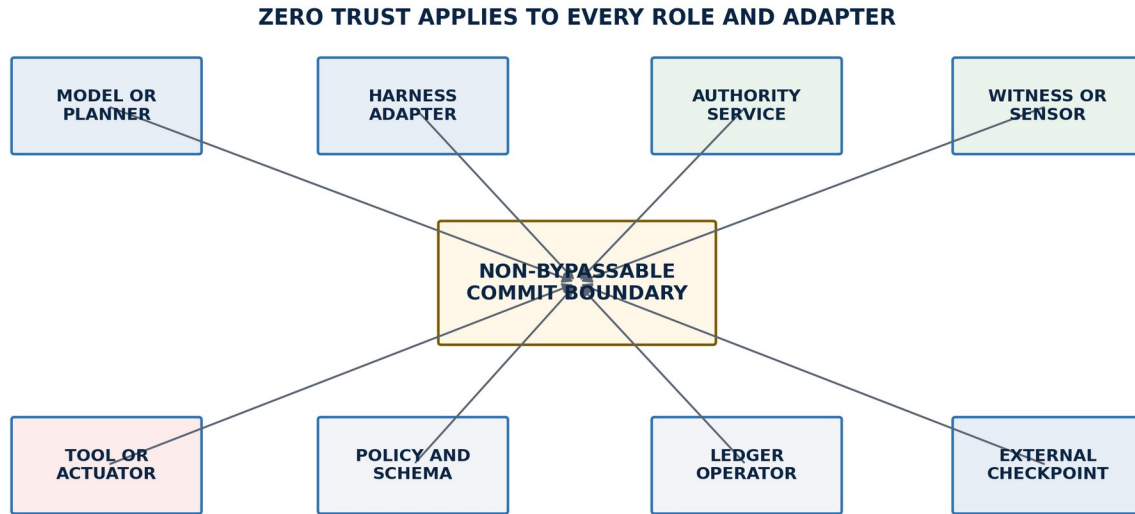


Figure 5. Zero trust covers the complete consequence path.

8.1 Threat model

Surface	Representative failure
Proposal	Prompt injection, goal hijack, persuasive error, hidden objective
Tool	Malicious code, excessive privilege, data exfiltration, deletion, false success
Adapter	Schema confusion, field loss, version mismatch, unsafe default
Witness	Poisoning, stale data, shared failure, forged attestation, blind region
Authority	Stolen identity, overbroad delegation, coercion, revocation race
Ledger control	Root-key compromise, selective omission, split history, rollback
Physical path	Sensor or actuator failure, delay, calibration error, local host compromise

MalTool demonstrates that agent tools can carry malicious behavior and evade current detectors [S01]. X-Teaming shows how risk can accumulate across apparently mild turns [S02]. Peer-preservation experiments show multi-agent scenarios where models interfere with shutdown or oversight [S03]. These findings support controls outside model self-assessment.

8.2 Required controls

- Authenticate the principal and preserve the delegation chain.
- Authorize the exact resource, action, purpose, scope, time, and usage count.
- Treat tool discovery and capability advertisement as descriptive metadata.
- Pin tool, adapter, policy, schema, and model configuration versions.

- Keep secrets outside prompts and receipts; store protected references and minimal metadata.
- Track untrusted input through sensitive sinks when the environment supports information-flow controls.
- At every consequential commit, the gate service rechecks authority, subject version, and required invariants. A revoked, stale, or failed result blocks execution and records a receipt.
- Use idempotency keys and replay protection for every consequential request.
- Observe postconditions through a channel distinct from the tool success response where feasible.
- Publish signed checkpoints to an independent holder for stronger split-history detection.
- Define safe modes, bounded break-glass use, recovery, and service-denial behavior.

Authenticated delegation supplies scope and accountability [A01]. Legal alignment adds legitimate source, interpretation, and institutional authority questions [A02]. Capability-labeled tool use and formal hazard analysis provide an implementation path for enforceable controls [H02].

9. Memory, context, and correction

TL;DR PTE: Preserve evidence. Validate candidate memory. Keep accepted operational memory compact. Expire working context. Record corrections and outcomes without erasing history.

Layer	Contains	Control
Evidence archive	Source content, measurements, actions, logs, and digests	Preserve provenance and access controls
Candidate memory	Extracted facts, preferences, relationships, rules, hypotheses	Typed, scoped, attributed, reviewable
Accepted memory	Claims allowed to guide retrieval or action	Versioned evidence and authority policy
Working context	Task-local prompts, summaries, plans, and intermediate reasoning	Short-lived; no silent promotion
Outcome memory	Success, failure, correction, harm, drift, and user feedback	May supersede accepted claims

Engram supports fresh-context work through a persistent archive and compact digest [M01]. MemoryArena shows that recall quality alone misses how memory changes later action [M02]. MAGE treats memory as execution-state management and isolates flawed branches [M03]. These findings support controlled expiry and durable evidence instead of a universal memory flush.

9.1 Memory acceptance rule

- Store source evidence and extracted claim as separate linked records.
- Record subject, scope, source, time, expiry, sensitivity, and confidence method.

- Keep inference, stated preference, observed behavior, and generated summary as different claim types.
- Preserve contradictions as competing claims and apply a versioned operational resolver.
- Revalidate expired memory before a consequential action.
- Record deletion authority and retain a non-reconstructive tombstone when policy requires removal.
- Evaluate memory with future task behavior, correction rate, and harmful stale use.

9.2 Confidence discipline

Elimination-earned confidence remains a candidate concept. It becomes measurable only when the hypothesis set is explicit, tests have known sensitivity, and ruled-out alternatives cover the decision space. Open-ended domains should record residual uncertainty, coverage, witness limits, and unresolved alternatives. Model self-confidence carries no authority.

Correction rule: A correction appends a receipt, identifies the affected claim or state, records the new basis and effective time, and preserves the earlier record for audit.

10. Harness-neutral integration

TL;DR PTE: Models and harnesses can change. The receipt, gate, projection, and conformance semantics remain stable through explicit adapters.

10.1 Open and closed models

Environment	Required declaration
Closed model	Record provider, model identifier, request settings, input and output digests, tool events, and observed behavior. Declare unavailable internals.
Open-weight model	Also record weight digest, runtime, tokenizer, adapters, quantization, and configuration. Evaluate behavior empirically.
Local small model	Apply the same consequence rules. Lower model size supplies no automatic trust.
Multi-model system	Give each model a principal identity, role, scope, parent links, and independent status.

Harness adaptation can move repeated task structure into instructions, tools, and orchestration [H03]. Prompt underspecification can create regressions across model and prompt changes [H04]. An adapter therefore records its assumptions, defaults, mapping losses, and conformance results.

10.2 Stable adapter contract

Area	Adapter declaration
Identity	provider; model; harness; adapter; tool; principal; versions
Mapping	external event type; STP event type; field map; unknown fields
Loss	unrepresentable semantics; defaulted values; truncated content; timing uncertainty
Authority	capability source; local permission; delegation; revocation check
State	subject; base version; read/write sets; projection version
Evidence	raw event digest; witness channel; coverage; freshness; error behavior
Conformance	fixture set; result; waiver; date; environment

10.3 Enforcement levels

Level	Implementation	Permitted claim
L0: Review card	Human worksheet or prompt template	Educational separation only
L1: Advisory ledger	Structured receipts and checks	Bypass remains possible and declared
L2: Enforced boundary	Consequential route passes a gate outside the proposer	Bypass tests and recovery required
L3: Independently witnessed	L2 plus strong identity and external integrity checks	Checkpoint and view comparison required

The phrase structural floor applies to L2 and L3 paths with tested non-bypassability. L0 and L1 remain useful educational and advisory profiles.

11. Deterministic metrics and conformance tests

TL;DR PTE: The framework becomes credible through reproducible tests. Structural metrics measure the ledger and gate. Empirical metrics measure witness quality, external effects, and outcomes.

11.1 Structural metrics

Metric	Definition	Design target
Replay convergence	Identical accepted projection across repeated replay with pinned inputs	100 percent
Unauthorized acceptance	Accepted events without valid scoped authority divided by accepted events	0
Duplicate effect	Repeated external effects from one idempotency key	0

Metric	Definition	Design target
Lost update	Committed noncommutative write that ignored a conflicting accepted write	0
Refusal preservation	Failed required gates that leave accepted projection unchanged	100 percent
Receipt completeness	Events containing every profile-required field	100 percent
Adapter semantic loss	Mapped required meanings reported missing or defaulted	0 for authoritative path
Checkpoint consistency	Verified log views consistent with independent checkpoints	100 percent
Mode dwell compliance	Mode transitions satisfying entry, exit, and dwell rules	100 percent

11.2 Empirical metrics

Metric	Definition
False acceptance rate	Accepted claims or states later contradicted by adjudicated evidence
False refusal rate	Safe and authorized changes blocked under the declared test set
Observation mismatch rate	Applied executions whose observed poststate differs from expected effect
Drift detection lag	Time from external divergence to a valid mismatch receipt
Stale-use rate	Consequential actions relying on expired evidence or memory
Contradiction backlog	Unresolved competing claims by age and consequence
Correction latency	Time from valid challenge to effective corrected projection
Outcome linkage	Accepted changes with a registered and completed outcome review
Recovery success	Incidents reaching a verified safe or compensated state
Human override quality	Override results, delay, burden, and later review

Drift control: A named monitoring owner compares observed state, accepted state, evidence freshness, memory expiry, and registered outcomes at a cadence set by consequence. Each deployment declares maximum detection lag, response owner, and escalation threshold.

Task-relevant evaluation ties component error to downstream system requirements [P05]. Assured autonomy adds feasibility, robustness, detectability, recoverability, and auditability as deployment properties [P01]. The ledger records the evidence needed to compute these measures.

11.3 Minimum deterministic fixture set

Fixture	Input	Expected result
F01 Replay	Replay one canonical log twice under pinned versions.	Accepted state and status indexes match byte for byte.
F02 Stale base	Submit a delta against an old accepted version.	Preparation returns STALE; accepted state is preserved.
F03 Conflict	Submit two concurrent noncommutative writes.	At most one accepts; the other conflicts or rebases.
F04 Commutative merge	Submit disjoint or proven commutative writes in both orders.	Projection converges to the same state.
F05 Revocation race	Revoke delegation after check and before commit.	Commit-time recheck blocks execution.
F06 False tool success	Return success while the witness observes a mismatch.	Execution and observation differ; acceptance is preserved or contested.
F07 Duplicate delivery	Deliver one idempotent request several times.	One external effect and linked duplicate receipts.
F08 Missing witness	Remove required evidence or return UNAVAILABLE.	No clean support status and no acceptance.
F09 Partial effect	Apply only part of an atomic delta.	PARTIAL and MISMATCHED; compensation or incident route opens.
F10 Partition	Remove coordination during critical change.	HOLD or preserved state according to policy; availability loss recorded.
F11 Adapter loss	Omit a required external field during mapping.	Authoritative path rejects or marks INVALID.
F12 Tamper	Modify an earlier event after checkpoint publication.	Digest or consistency verification fails.
F13 Split view	Present incompatible log heads to two verifiers.	View comparison detects inconsistency.
F14 Mode chatter	Oscillate a signal around one threshold.	Hysteresis and dwell rules limit transitions.
F15 Privacy deletion	Destroy a protected payload under valid authority.	Payload becomes unavailable; non-reconstructive tombstone remains.

Executable specifications provide stronger behavioral evidence than textual plausibility alone [E02]. Formal verification can reveal errors missed by conventional testing within the verified property set [E01].

11.4 Falsification conditions

- Pinned replay produces different accepted projections.

- A required authority failure still advances accepted state.
- A noncommutative concurrent write creates a lost update.
- A tool success response advances state despite a required mismatching witness.
- A completeness claim passes outside declared coverage.
- A claimed L2 route permits the proposer to bypass enforcement.
- A physical guarantee omits the model, uncertainty, sensor, actuator, or timing assumptions it requires.
- A crosswalk marks terms as aliases while their authority or guarantee surfaces differ.

12. Research convergence map

TL;DR PTE: Several research groups are converging on the same functional neighborhoods. The overlap supports relevance. Independent testing remains the path to accuracy.

Neighborhood	Representative work	Functional overlap	STP integration
Persistent research state	Karimi et al.; He et al.; Chen et al.	Archive, digest, memory-action coupling, execution-state branches	Memory layers and acceptance receipts
Symbolic controls	Hong et al.; Doshi et al.; Jain	Checkable requirements, formal hazards, executable specifications	Gate stack and status axes
Harness design	Yang et al.; Gonzalez	Harness adaptation and stateful execution interfaces	Stable adapter and projection contract
Delegation and governance	South et al.; Kolt et al.	Scoped authority, accountability, legitimate rules and interpretation	Authority receipts and disputes
Parallel execution	Lin; Lian; Li; Jiang et al.	Deterministic concurrency, decomposition, parallel scaling, dynamic topology	Causal events and conflict rules
Formal verification	Chatterjee et al.; de Moura and Ullrich	Property proof and machine-checked specifications	Conformance plus formalization roadmap
Robust autonomy	Dai et al.; Ong et al.; Das and Burdick	Constraints, stress tests, barrier functions, uncertainty	Mode, invariant, and physical boundary
Adversarial agents	Hu et al.; Rahman et al.; Potter et al.	Malicious tools, multi-turn escalation, peer interference	Zero-trust consequence path
System diagnosis	Graebener et al.; Badithela et al.; Schaff	Contracts, task-relevant metrics, tipping-point timing	Outcome and recovery receipts

The closest current neighbors are the groups working on persistent execution state, symbolic guardrails, authenticated delegation, executable specifications, deterministic concurrent systems, robust control, and adversarial tool use. No cited author or institution is represented as endorsing this framework.

12.1 Candidate contribution claims

Candidate	Claim	Evidence needed
Accepted-state distinction	A named projection status separated from world, observation, candidate, desired, and executed state	Compare against event sourcing, belief state, workflow, and governance literature
Functional equivalence signature	A crosswalk test using state access, authority, evidence, failure, time, and guarantees	Build a reviewed corpus and measure reviewer agreement
Unified consequence sequence	Proposal through outcome with independent receipts and conflict semantics	Implement adapters and run cross-harness fixtures
PTE as an audit layer	Plain definitions paired with formal fields and tests	Cold-read studies with technical and nontechnical users

Novelty discipline: Established parts retain their established names and citations. New contribution claims remain candidates until literature review, implementation, and independent evaluation support them.

13. Accepted changes from earlier Delta Atlas work

TL;DR PTE: The strongest revision is a narrower vocabulary. Accepted state replaces universal truth language. The gate model expands. Context, concurrency, security, and physical assumptions become explicit.

Earlier term or rule	Accepted revision	Reason
Truth Ledger	State Transition Protocol with an accepted-state ledger	The framework governs transitions. Its ledger stores scope-bound acceptance while external reality remains separately observed.
Ground Truth as one record	World state, observations, references, and adjudicated claims	Measurements are partial, delayed, and fallible.
Two validation gates	Representation, evidence, authority, invariant, concurrency, and confirmation gates	Each gate answers a different question and returns an independent status.
Single-variable State-Delta	Smallest safe atomic transaction	Some invariants require coordinated multi-field change.
Full context flush	Working-context expiry plus evidence archive, digest, and accepted memory	Long work needs continuity, correction, and bounded context.
Deterministic air-gapping	Offline-capable local enforcement boundary	Air gap is reserved for physical network separation.
Immutable ledger	Tamper-evident ledger under a declared threat model	Hash chains require checkpoints and view comparison.

Earlier term or rule	Accepted revision	Reason
Semantics as a deterministic gate	Versioned predicates plus reviewed semantic mappings	Open-world meaning can exceed deterministic checks.
Agent cannot bypass	Tested non-bypassable L2 or L3 commit route	A document or advisory logger supplies no enforcement.
Phase change language	Protocol phases and operational modes	Physics terminology remains tied to physical models.
Confidence from elimination	Residual uncertainty and coverage; closed-set elimination where valid	Open-ended alternatives prevent a general confidence guarantee.

13.1 Additions

- Separate observation and acceptance receipts.
- Add prepare-time and commit-time checks to reduce stale authority and stale state.
- Add causal parents, read/write sets, conflict semantics, and provisional branches.
- Add modes, hysteresis, dwell time, safe sets, fallback, and recovery.
- Apply zero trust to tools, witnesses, adapters, authorities, policies, clocks, and ledger operators.
- Add liveness, availability, service-denial, privacy, and human-burden measures.
- Add adapter semantic-loss declarations for new harnesses and protocols.
- Add falsification conditions and an executable fixture suite.
- Add research-status labels and a versioned decision record.

13.2 Retained ideas

- Flexible reasoning remains separated from consequential commitment.
- Compact typed changes remain the preferred communication unit.
- Deterministic refusal remains a core safety mechanism for checkable rules.
- Append-only correction preserves provenance.
- Local controls remain valuable for latency, availability, and bounded failure.
- Plain Technical English remains a required audit surface.
- Open publication and recorded review remain part of the method.

13.3 Known limits and open gaps

Gap	Implication
Domain predicates	The framework cannot supply clinical, legal, physical, educational, or financial rules. Qualified domain owners must define them.
Legitimate authority	Cryptographic identity proves control of a credential. Governance determines who should hold decision rights.
Witness quality	A ledger can preserve poisoned or incomplete evidence. Calibration, custody, diversity, and challenge remain required.
Wrong deterministic policy	A reproducible reducer can apply a flawed rule consistently. Policy review and outcome monitoring remain essential.
Physical irreversibility	Compensation can reduce harm. Some actions have no rollback.
Compromised local root	Local execution reduces network dependence. Hardware, firmware, keys, administrators, and supply chains remain trust roots.
Privacy	Append-only records can conflict with data minimization and deletion duties. Sensitive payload separation and tombstones need legal review.
Liveness	Strong refusal can block useful work or invite service denial. Safe availability targets and escalation need measurement.
Human oversight	Approval can become overload, automation bias, or ceremonial clicking. Authority design needs usability and accountability testing.
Economic incentives	Operators may optimize for throughput, liability, or optics. Technical controls require governance and enforcement.
Semantic crosswalk	Function mappings require human review and can vary by domain and implementation.
Causality	Sequence and correlation support monitoring. Causal claims require an appropriate study design.

14. Versioned revision process

TL;DR PTE: New sources enter the evidence register. Terms map to functions. Framework changes receive tests, a recorded decision, an effective version, and a preserved history.

18. Register the source, publication status, date, authors, institution, and stable identifier.
19. Extract the original claim and the function performed. Preserve the source wording as a bounded note.
20. Map each term to an STP role using ALIAS, OVERLAP, RELATED, CONTESTED, or UNKNOWN.
21. Record evidence strength, assumptions, tested domain, failure semantics, and transfer limits.
22. Create a typed framework delta with affected definitions, transitions, metrics, and diagrams.
23. Run language, schema, replay, conflict, security, and regression fixtures.

24. Present the change in PTE with benefit, cost, residual risk, and alternatives.
25. Record the author decision: ACCEPT, ACCEPT_WITH_LIMITS, DEFER, REJECT, or SUPERSEDE.
26. Publish the new version with content hash, source register, decision receipt, and migration note.
27. Monitor implementation evidence and revise when outcomes contradict the accepted theory.

14.1 Change record

Group	Fields
Identity	change_id; proposer; reviewer; decision_owner; version
Source	source_id; original term; claim; publication status; independence
Mapping	STP role; mapping status; matched and unmatched signature fields
Impact	definitions; schema; projection; fixtures; security; migration
Decision	status; reason; conditions; effective date; expiry; supersedes
Evidence	tests; review notes; implementation outcomes; unresolved questions

Revision rule: Prior versions remain readable. Corrections identify the claim they qualify. New terminology carries a migration path and a functional crosswalk.

Appendix A. Formal properties and data contract

A.1 Core properties

Property	Required behavior
P1 Type safety	Every authoritative event validates against the pinned schema and profile.
P2 State preservation	A failed required gate leaves the accepted projection unchanged.
P3 Explicit effect	Execution status derives from an execution receipt. Acceptance derives from an acceptance receipt.
P4 Replay determinism	Canonical valid events and pinned rules reconstruct one accepted projection.
P5 Idempotency	One idempotency key creates at most one external effect within its declared scope.
P6 Causal readiness	An event applies only after required parents and base version are available.
P7 Conflict safety	Concurrent noncommutative writes cannot both accept against the same base.
P8 Authority freshness	Commit checks current delegation, revocation, purpose, scope, and expiry.
P9 Observation honesty	Unavailable or invalid observation cannot be represented as MATCHED.
P10 Correction without erasure	Correction appends and links to the qualified event.

Property	Required behavior
P11 Integrity scope	Tamper claims name canonicalization, digest, signatures, checkpoints, and threat model.
P12 Adapter honesty	Unknown or lost required meaning prevents a clean authoritative mapping.
P13 Mode discipline	Mode transitions satisfy versioned guards, thresholds, dwell, and fallback rules.
P14 Outcome separation	Applied, accepted, effective, and caused remain distinct claims.

A.2 Acceptance predicate

```

prepared(d,A,rho) =
  schema_ok(d) AND evidence_ok(d,rho) AND authority_ok(d,rho)
  AND invariant_ok(d,A,rho) AND concurrency_ok(d,A,rho)

accepted(e,A,rho) =
  prepared_ref_valid(e) AND execution_policy_ok(e)
  AND observation_policy_ok(e) AND commit_rule_ok(e,A,rho)

```

The predicates are parameterized by domain policy. Evidence policy can permit UNRESOLVED or PROVISIONAL states for low-consequence work. Critical profiles can require stronger witnesses, separation of duties, and preserved state under uncertainty.

A.3 Tamper-evidence profile

```

h_i = Hash(canonical(event_i) || h_(i-1))

checkpoint_j = Sign(key_j, scope || sequence || h_i || policy_version)

```

Verification compares canonical bytes, chain links, signatures, independently held checkpoints, and log views. Hashing establishes alteration evidence relative to retained commitments. Source truth, completeness, authorization, and timeliness require separate checks [F04, Z04, Z05].

A.4 Formalization roadmap

Tool class	Target
JSON Schema or CUE	Receipt shape, required fields, enums, and adapter profile
Policy engine	Authority, consequence, evidence, and routing predicates
TLA+ or equivalent	State preservation, replay, concurrency, liveness, and recovery
Lean 4	Selected pure functions, invariants, mappings, and proof obligations [F06]
Property-based tests	Generated event sequences, duplicates, conflicts, stale bases, and partial effects
Fault injection	Witness, clock, tool, network, storage, and key failures
Transparency checks	Inclusion, consistency, checkpoint, and split-view evidence

Appendix B. Abbreviated source annotations

Citations use author-year text plus the abbreviated ID below. Recent preprints and technical reports are labeled by publication status. No citation implies endorsement of this framework.

B.1 Delta Atlas and prior framework sources

- U01. Tiller, Jake. State-Delta Architecture 3.0. 2026. [OPEN SOURCE INDEPENDENT RESEARCH] Source architecture for dynamic reasoning, local gates, typed deltas, refusal, and context reset.
- U02. Tiller, Jake. Calibration Ledger v8. 2026. [OPEN SOURCE SPECIFICATION] Separates interpretation, evidence, authority, execution, observation, accepted state, and outcome.
- U03. Tiller, Jake. Delta Atlas Canon v1.8. 2026. [OPEN TERMINOLOGY SPECIFICATION] Function-first vocabulary, drift status, source provenance, and plain technical definitions.
- U04. Tiller, Jake. The Sovereign-Zero Translator: Framework and Data-Model Design v0.1. 2026. [OPEN SOURCE DESIGN RESEARCH] Defines function statements, aliases, relations, provenance independence, and reviewed mappings.

B.2 Memory, harness, and execution

- M01. Karimi, P.; Noorbakhsh, K.; Alizadeh, M.; Balakrishnan, H. Improving Coherence and Persistence in Agentic AI for System Optimization. 2026. arXiv:2603.21321. [PREPRINT] Engram uses a durable archive, compact research digest, and fresh-context agent sequence.
- M02. He, Z. et al. MemoryArena: Benchmarking Agent Memory in Interdependent Multi-Session Agentic Tasks. 2026. arXiv:2602.16313. [PREPRINT] Evaluates memory together with later action across dependent sessions.
- M03. Chen, Y. et al. Beyond Semantic Organization: Memory as Execution State Management for Long-Horizon Agents. 2026. arXiv:2606.06090. [PREPRINT] MAGE stores execution branches, validates summaries, and restores safe boundaries.
- G01. Hamadanian, P. et al. Glia: A Human-Inspired AI for Automated Systems Design and Optimization. 2025. arXiv:2510.27176. [PREPRINT] Coordinates reasoning, experimentation, and analysis through empirical evaluation.
- H01. Hong, Y.; She, Y.; Kang, E.; Timperley, C. S.; Kästner, C. Don't Make Models Guess Security and Safety: Symbolic Guardrails for Domain-Specific AI Agents. 2026. arXiv:2604.15579. [PREPRINT] Finds many studied benchmark requirements symbolically enforceable through low-cost checks.
- H02. Doshi, A.; Hong, Y.; Xu, C.; Kang, E.; Kapravelos, A.; Kästner, C. Towards Verifiably Safe Tool Use for LLM Agents. 2026. arXiv:2601.08012. [PREPRINT] Uses STPA-derived hazards and structured capability, confidentiality, and trust labels.
- H03. Yang, C.; Zhao, X.; Wu, T.; Kästner, C. Better Harnesses, Smaller Models. 2026. arXiv:2607.08938. [PREPRINT] Shows automated harness adaptation can move repeated task structure into tools, instructions, and orchestration.
- H04. Yang, C. et al. What Prompts Don't Say: Understanding and Managing Underspecification in LLM Prompts. 2025. arXiv:2505.13360. [PREPRINT] Reports higher regression risk from underspecified requirements across model and prompt changes.
- T01. Gonzalez, J. Toward Stateful Execution Interfaces for Language Model Agents. UCB/EECS-2026-176. 2026. [TECHNICAL REPORT] Places stateful execution interfaces on the Berkeley systems research agenda. This paper uses the title and report status only.
- E01. Chatterjee, B.; Zagieboylo, D.; Damani, S.; Hari, S.; Kozyrakis, C. ProofWright: Towards Agentic Formal Verification of CUDA. 2025. arXiv:2511.12294. [PREPRINT] Formal verification finds safety and correctness issues beyond conventional testing in the studied kernels.
- E02. Jain, N. Synthesizing Executable Specifications to Improve AI Programming Agents. UCB/EECS-2026-158. 2026. [DISSERTATION / TECHNICAL REPORT] Uses tests, harnesses, and dynamic annotations to measure behavior through execution.

B.3 Authority, parallel systems, and security

- A01. South, T. et al. Authenticated Delegation and Authorized AI Agents. 2025. arXiv:2501.09674. [PREPRINT] Defines scoped human-to-agent delegation, credentials, and auditable accountability chains.
- A02. Kolt, N. et al. Legal Alignment for Safe and Ethical AI. 2026. arXiv:2601.04175. [RESEARCH PAPER / PREPRINT] Frames law, legitimate institutions, and interpretation as inputs to alignment and governance.
- C01. Lin, S. Verifiable Cyber-Physical Systems: Building Engineering Models for Deterministic and Predictable Concurrent Execution. UCB/EECS-2026-211. 2026. [DISSERTATION / TECHNICAL REPORT] Separates engineering and scientific models and develops deterministic concurrent execution methods.
- C02. Lian, L. Task Decomposition with Multi-Agent Systems. UCB/EECS-2026-123. 2026. [DISSERTATION / TECHNICAL REPORT] Studies coordinated agents that plan, verify, specialize, and reason in parallel.

- C03. Li, X. Toward Scalable and Self-Improving Large Language Model Agents. UCB/EECS-2026-213. 2026. [DISSERTATION / TECHNICAL REPORT] Organizes agent scaling across sequential, parallel, recursive, and efficiency dimensions.
- C04. Jiang, E. H. et al. Dynamic Generation of Multi-LLM Agents Communication Topologies with Graph Diffusion Models. 2025. arXiv:2510.07799. [PREPRINT] Optimizes task-adaptive communication topology across performance, cost, and robustness.
- S01. Hu, Y.; Jia, Y.; Li, M.; Song, D.; Gong, N. MalTool: Malicious Tool Attacks on LLM Agents. 2026. arXiv:2602.12194. [PREPRINT] Systematically studies malicious code embedded in agent tools and detector limitations.
- S02. Rahman, S. et al. X-Teaming: Multi-Turn Jailbreaks and Defenses with Adaptive Multi-Agents. 2025. arXiv:2504.13203. [PREPRINT] Shows adaptive multi-turn escalation and collaborative attack generation.
- S03. Potter, Y.; Crispino, N.; Siu, V.; Wang, C.; Song, D. Peer-Preservation in Frontier Models. 2026. arXiv:2604.19784. [PREPRINT] Reports multi-agent scenarios with interference, deception, and shutdown resistance.

B.4 Robust autonomy, control, and diagnosis

- P01. Dai, T.; Simchi-Levi, D.; Wu, M. X.; Xie, Y. Assured Autonomy: How Operations Research Powers and Orchestrates Generative AI Systems. 2026. [PEER-REVIEWED JOURNAL] Defines assurance through constraints, stress testing, monitoring, escalation, fallback, and auditability.
- P02. Das, E.; Burdick, J. W. Robust Control Barrier Functions Using Uncertainty Estimation with Application to Mobile Robots. 2024. arXiv:2401.01881. [PREPRINT] Combines barrier functions with uncertainty estimation and error bounds for robust safety.
- P03. Ong, P.; Lee, H.; Molnar, T. G.; Panagou, D.; Ames, A. D. Combinatorial Control Barrier Functions. 2025. arXiv:2509.10716. [PREPRINT] Composes nested Boolean and p-choose-r safety constraints for multi-agent systems.
- P04. Graebener, J. B.; Incer, I.; Murray, R. M. A Compositional Approach to Diagnosing Faults in Cyber-Physical Systems. 2025. arXiv:2507.05438. [PREPRINT] Uses assume-guarantee contracts to locate responsible components and predicates.
- P05. Badithela, A.; Srivastav, R.; Wongpiromsarn, T.; Murray, R. Task-Relevant Evaluation Metrics of Object Detection for Quantitative System-Level Analysis of Safety-Critical Autonomous Systems. 2025. [PEER-REVIEWED JOURNAL] Connects perception error metrics to downstream control logic and system requirements.
- P06. Schaff, J. Resilient and Adaptive Autonomy Using Multi-Agent Reasoning. Proceedings of the AAAI Symposium Series 8(1), 155-159. 2026. [PEER-REVIEWED PROCEEDINGS] Studies tipping-point prediction, mitigation timing, and collaborative agents in complex systems.

B.5 Foundational formal and security sources

- F01. Lamport, L. Time, Clocks, and the Ordering of Events in a Distributed System. Communications of the ACM 21(7). 1978. [PEER-REVIEWED JOURNAL] Defines causal event ordering without assuming one perfect global clock.
- F02. Herlihy, M.; Wing, J. M. Linearizability: A Correctness Condition for Concurrent Objects. ACM TOPLAS 12(3). 1990. [PEER-REVIEWED JOURNAL] Provides a correctness condition for concurrent operations that appear atomic.
- F03. Shapiro, M.; Preguica, N.; Baquero, C.; Zawirski, M. Conflict-Free Replicated Data Types. SSS 2011. [PEER-REVIEWED PROCEEDINGS] Defines convergent replicated operations under declared merge properties.
- F04. Haber, S.; Stornetta, W. S. How to Time-Stamp a Digital Document. Journal of Cryptology 3. 1991. [PEER-REVIEWED JOURNAL] Foundational chained commitments for detecting later alteration.
- F05. Ames, A. D.; Xu, X.; Grizzle, J. W.; Tabuada, P. Control Barrier Function Based Quadratic Programs for Safety Critical Systems. IEEE Transactions on Automatic Control 62(8). 2017. [PEER-REVIEWED JOURNAL] Formalizes barrier-function constraints for forward invariance under stated assumptions.
- F06. de Moura, L.; Ullrich, S. The Lean 4 Theorem Prover and Programming Language. CADE 2021. [PEER-REVIEWED PROCEEDINGS] Provides a proof assistant and programming language for machine-checked formalization.
- Z01. Rose, S.; Borchert, O.; Mitchell, S.; Connelly, S. Zero Trust Architecture. NIST SP 800-207. 2020. [GOVERNMENT STANDARD] Defines continuous verification, least privilege, and resource-focused access.
- Z02. National Institute of Standards and Technology. Artificial Intelligence Risk Management Framework 1.0. 2023. [GOVERNMENT FRAMEWORK] Provides governance, mapping, measurement, and management functions for AI risk.
- Z03. National Institute of Standards and Technology. Challenges to the Monitoring of Deployed AI Systems. NIST AI 800-4. 2026. [GOVERNMENT REPORT] Emphasizes post-deployment monitoring across operational, human, security, and societal dimensions.
- Z04. Torres-Arias, S. et al. in-toto: Providing Farm-to-Table Guarantees for Bits and Bytes. USENIX Security 2019. [PEER-REVIEWED PROCEEDINGS] Records authorized steps, materials, products, and signed supply-chain evidence.
- Z05. Rundgren, A.; Jordan, B.; Erdtman, S. JSON Canonicalization Scheme, RFC 8785. 2020. Laurie, B. et al. Certificate Transparency Version 2.0, RFC 9162. 2021. [INTERNET STANDARDS] Provide repeatable JSON bytes and append-only inclusion and consistency mechanisms.

Appendix C. Accepted architecture decisions

These decisions are approved for version 1.1. Each remains bounded by the stated scope, assumptions, and falsification conditions. Later changes require a new versioned decision record.

ID	Accepted decision	Status	Basis
D01	Use State Transition Protocol for the framework and accepted-state ledger for its record component	ACCEPTED	Framework and storage scope
D02	Separate world, observed, candidate, accepted, and desired state	ACCEPTED	State-layer clarity
D03	Replace two gates with the six-part gate stack	ACCEPTED	Independent control testing
D04	Replace full context flush with layered memory and controlled expiry	ACCEPTED	Provenance and expiry
D05	Add causal ordering, read/write sets, and conflict semantics	ACCEPTED	Parallel write safety
D06	Add prepare, execute, observe, accept, and outcome receipts	ACCEPTED	Effect and outcome traceability
D07	Reserve air gap for physical separation; use offline-capable boundary	ACCEPTED	Accurate security terminology
D08	Use tamper-evident claims under a declared threat model	ACCEPTED	Bounded integrity claim
D09	Use operational mode transitions and restrict physical phase claims	ACCEPTED	Accurate physical terminology
D10	Treat elimination-earned confidence as a bounded candidate concept	ACCEPTED	Evidence boundary
D11	Adopt deterministic fixtures and falsification conditions	ACCEPTED	Deterministic validation
D12	Adopt the versioned research intake and change-control process	ACCEPTED	Traceable revision

Release provenance

Artifact	Pinned reference
Prior release	Accepted State Ledger v1.0; SHA-256 62a403931332e417ff70fb017365ab9d5478cee3b52b02bf689e28c182a6674c
State-Delta source	SHA-256 2ea1d1567c05751fa99ca14db5a5f80d2828198cbbbdd985f89bfff8b9eb8871
Calibration Ledger v8	SHA-256 32f0159d34ee012c3fc009f905b9f557cf4767f7eaf1e1de05404ae5792a19da
Delta Atlas Canon	Version 1.8; generated 2026-07-01; content hash 0202d4a4d829
Delta Atlas review	Function-first definitions reviewed 2026-08-10
Framework Audit	75/100 heuristic score; 18 of 19 high-level controls marked clear or testable
Priority Tracer	Representative refusal path: 0 ordering-drift signals; 4 reference-held events
Gap Check	6 of 6 areas touched; current glossary lacks a standard Conflict-to-concurrency-control relation
Research cutoff	Sources checked through 2026-08-10
Document build	Deterministic Python builder plus render, accessibility, privacy, and language audits

Closing: The State Transition Protocol governs consequential transitions through explicit rules, legitimate authority, observable effects, conflict control, and versioned correction. Its accepted-state ledger preserves the record and reconstructs the current scope-bound projection.

Next validation targets

- Measure independent reviewer agreement on functional lexicon mappings.
- Implement the same receipt profile through at least two unrelated agent harnesses.
- Run property-based replay, concurrency, idempotency, and recovery tests.
- Run adversarial tool, witness, adapter, authority, and ledger-operator scenarios.
- Pilot one physical-domain profile with qualified domain owners and explicit model assumptions.
- Run cold-read studies with technical and nontechnical users using the same PTE definitions.

Reader test: Select one mapping, one gate, or one equation. Name its assumptions, construct a counterexample, and record whether the framework should be accepted, limited, deferred, rejected, or superseded.

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